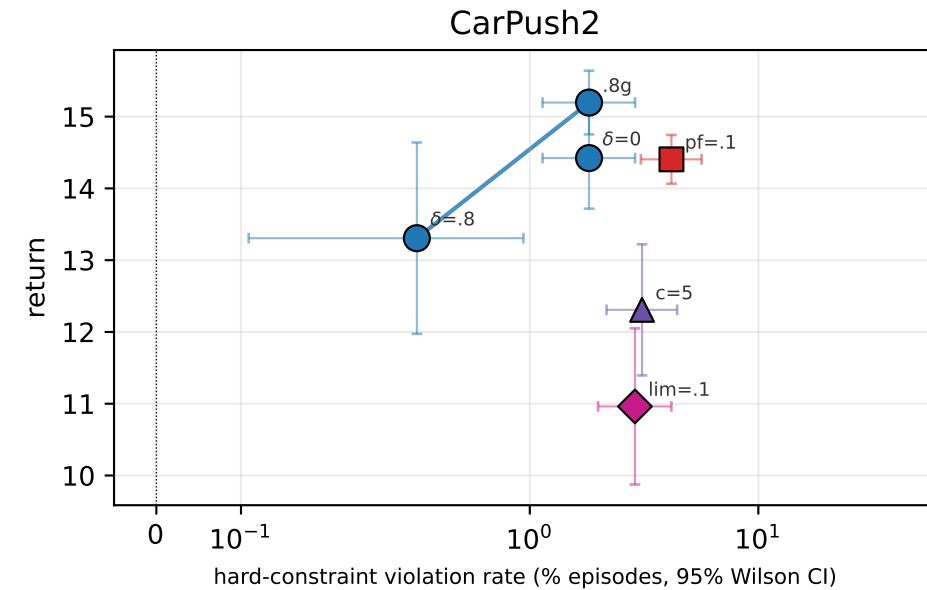
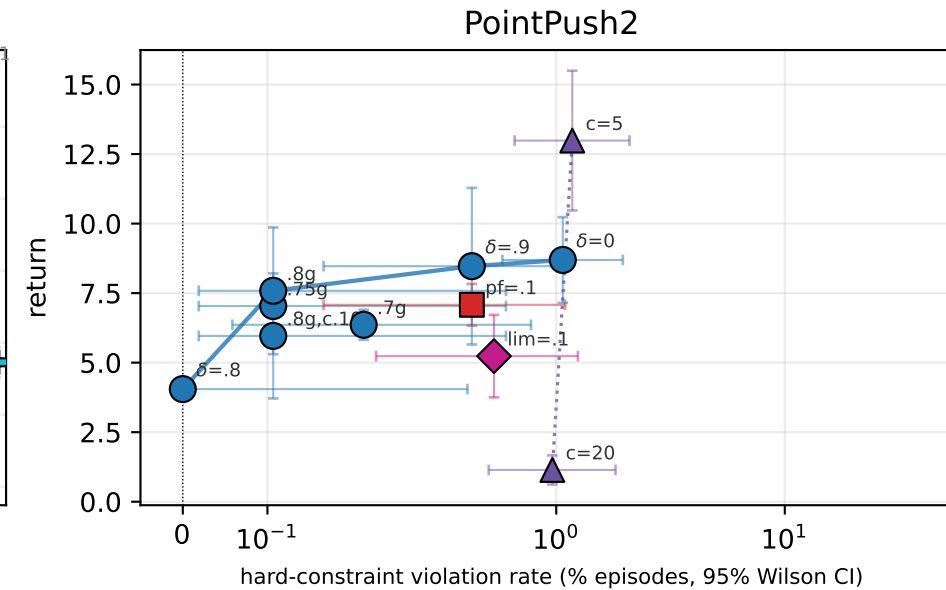
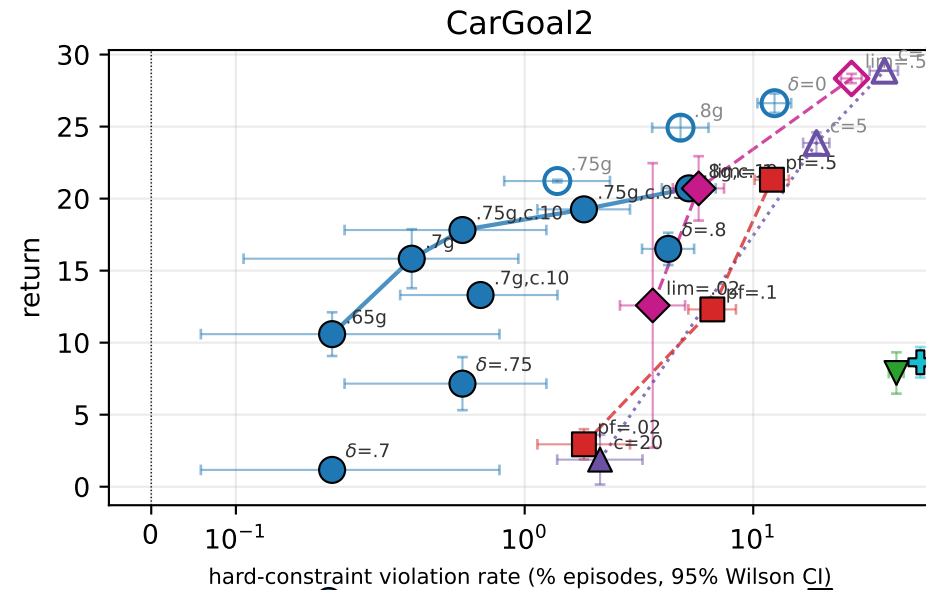


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



● SafeQT-typed (ours)

- safeOT typed (cars)
- FDDL / constraint based

FDPI (constraint-bas

◆ SAC-Lagrangian (pe

▲ SAC + fixed penalty

SDAC (multi-constraint SOTA)

 SDAC (multi-constraint SOTA)  
 SBCBO (multi-constraint SOTA)

- ✚ SRCPO (multi-constraint SOTA)

— typed Pareto frontier (soft  $\leq$ )

TA) ..... SAC-Pen coefficient

DTA) EDPL is sweeter

FDPI  $p_f$  sweep

≤ 25)      --- SAC-Lag cost-limit s

3011 Cl)

hard-constraint violation rate (%) e